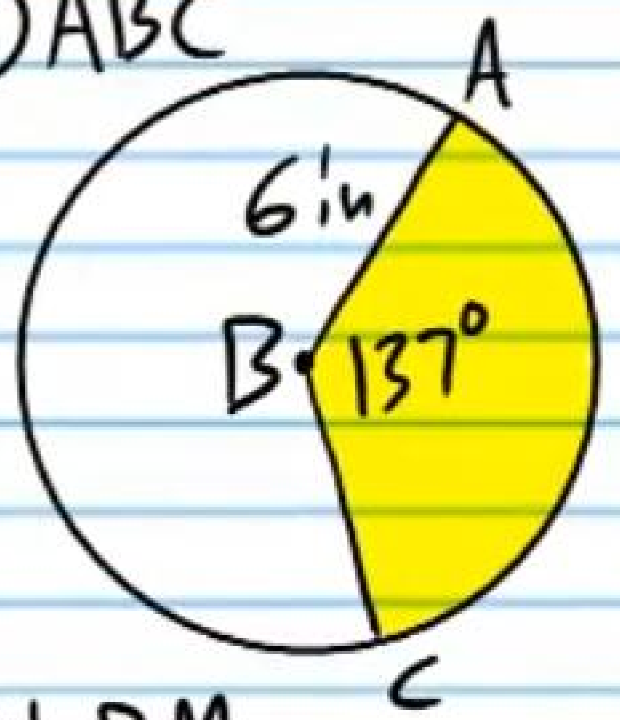
Name
Date
Course

Circles (Part II): Sectors, Chords, and Graphs of Circles (Homework)

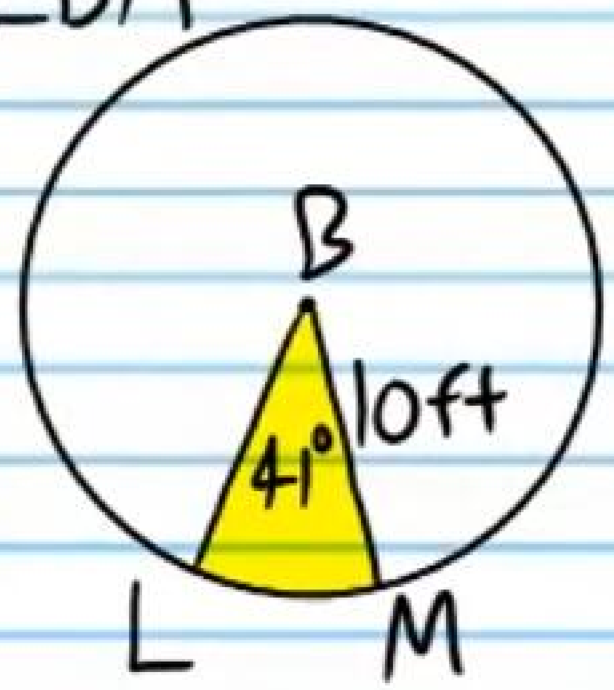
If B is the center of the following circles, find the area of the sectors below. You must use the π button on your calculator. Round to the nearest tenth.

① ABC



$\boxed{43\text{ in}^2}$

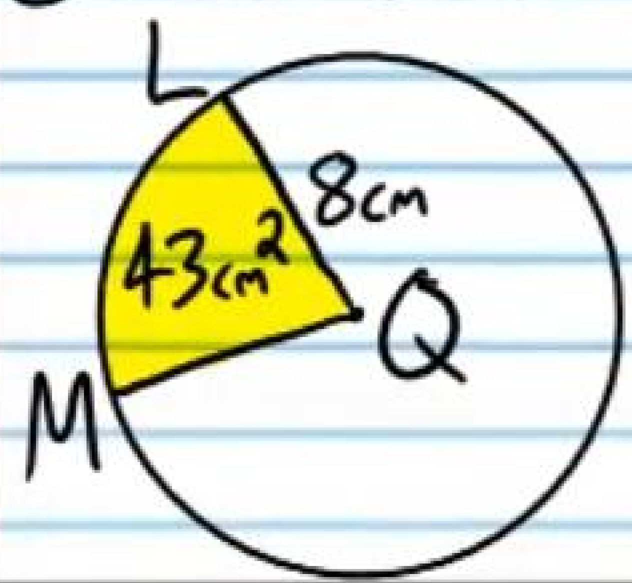
② LBM



$\boxed{35.8\text{ ft}^2}$

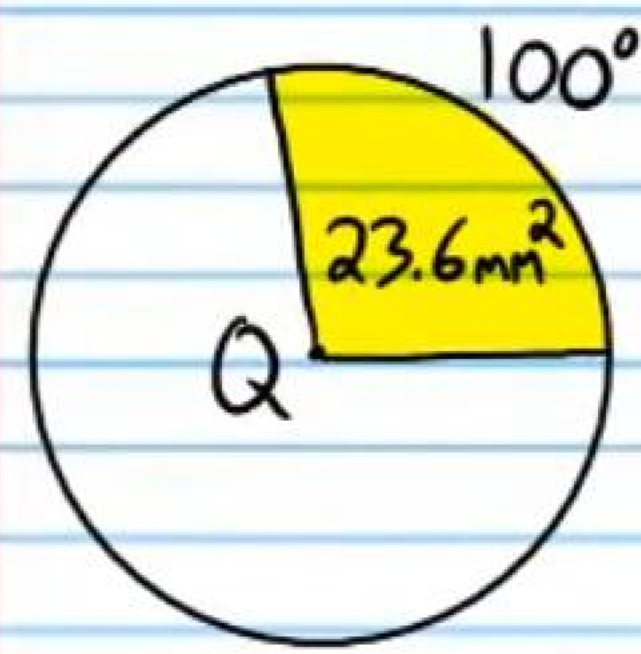
If Q is the center of the following circles, find the following values. You must use the π button on your calculator. Round to the nearest tenth.

③ $m\angle LQM$



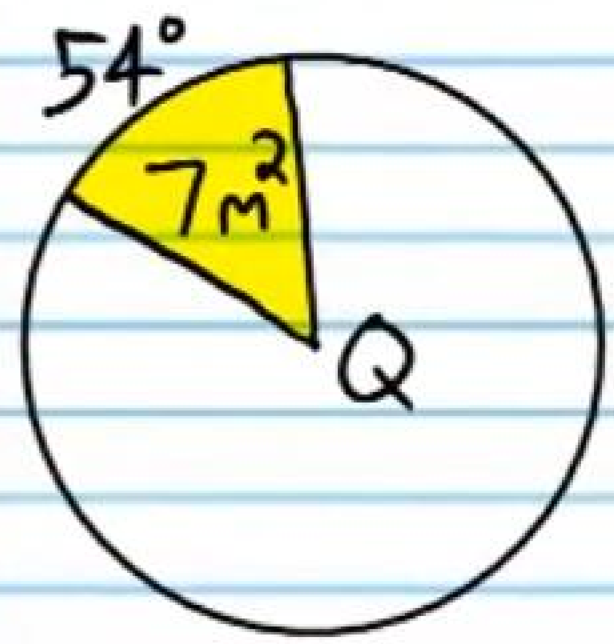
$\boxed{77^\circ}$

④ Radius



$\boxed{5.2\text{ mm}}$

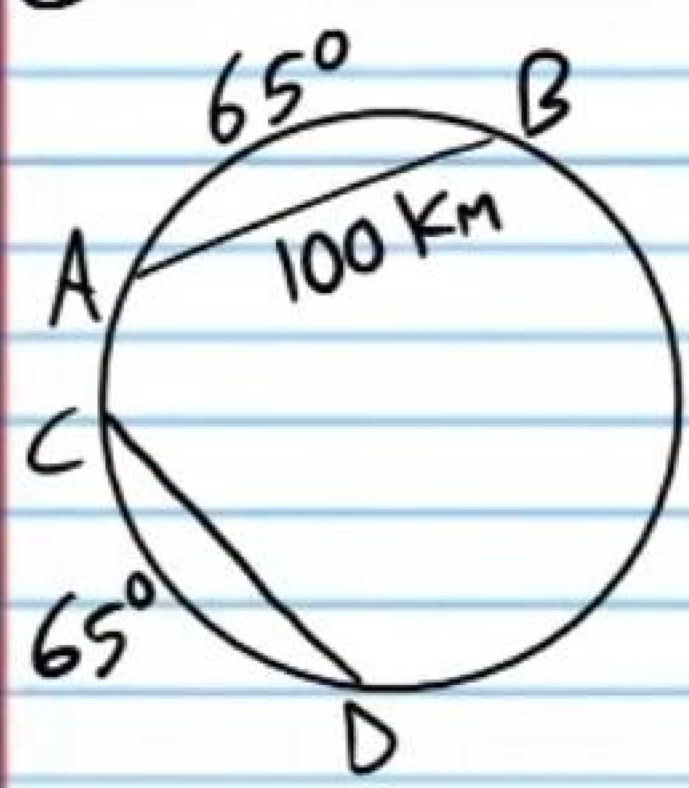
⑤ Area of ⊙Q



$\boxed{46.7\text{ m}^2}$

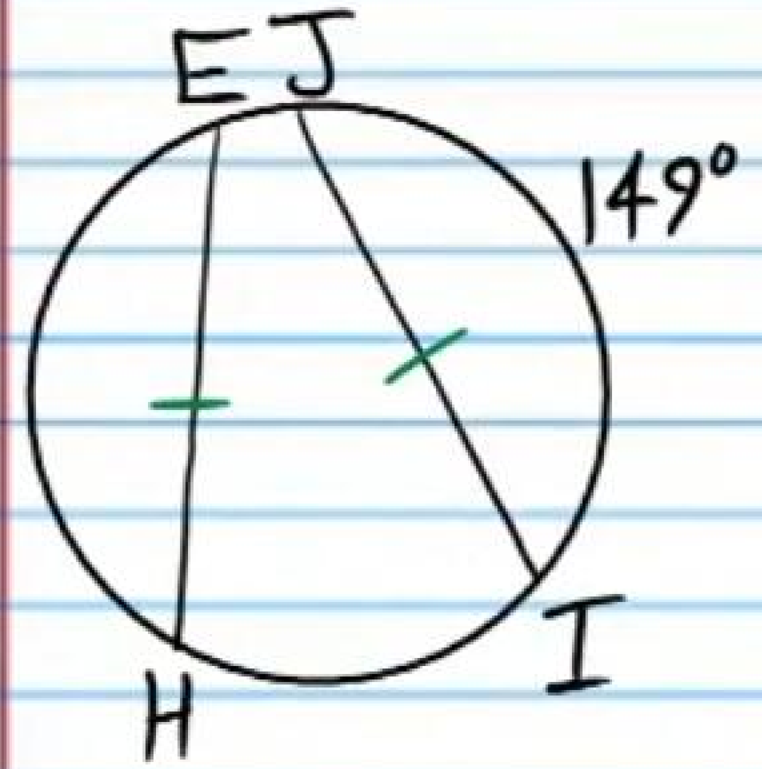
Find the following values.

⑥ CD



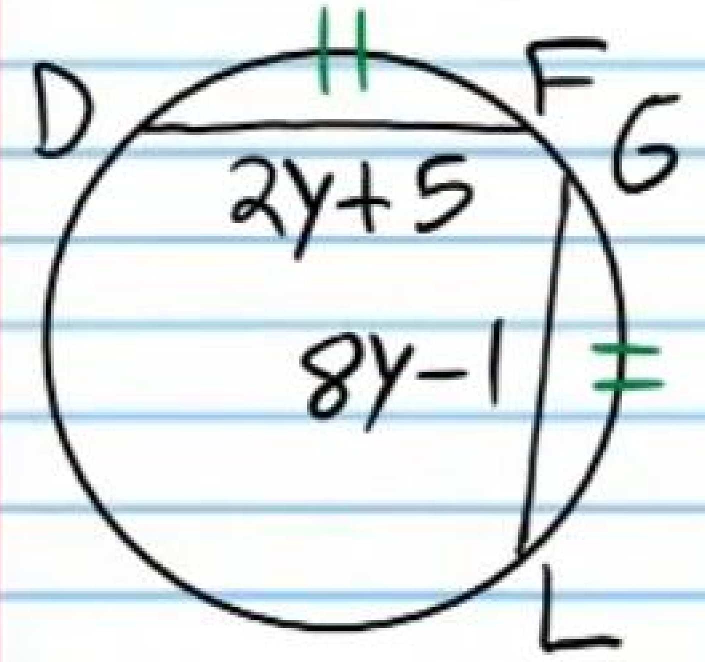
$\boxed{100\text{ Km}}$

7. $m\widehat{EH}$



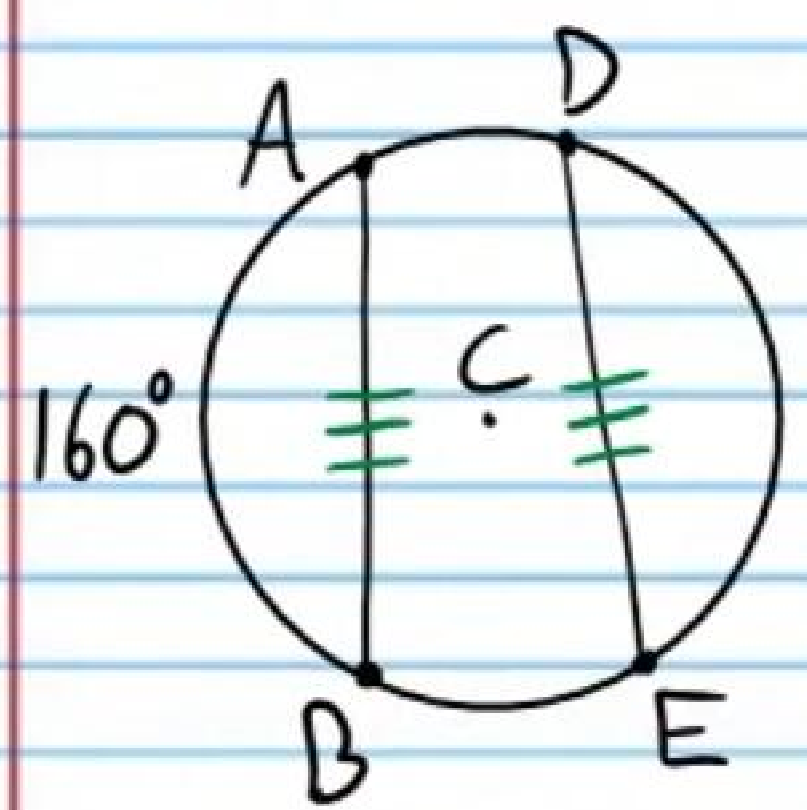
$\boxed{149^\circ}$

8. GL



$\boxed{7}$

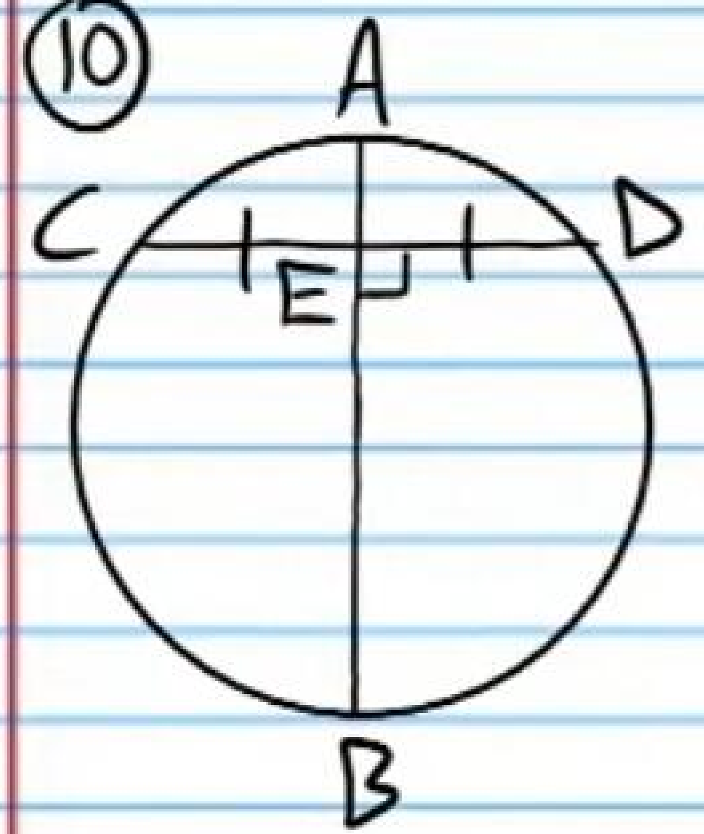
9. $m\angle DCE$



$\boxed{160^\circ}$

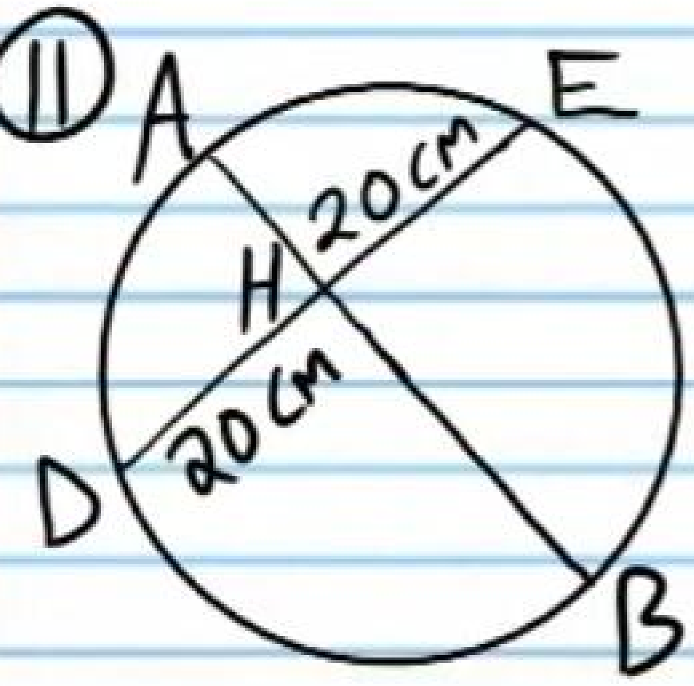
Is $\overline{AB}$ a diameter? Explain.

10.



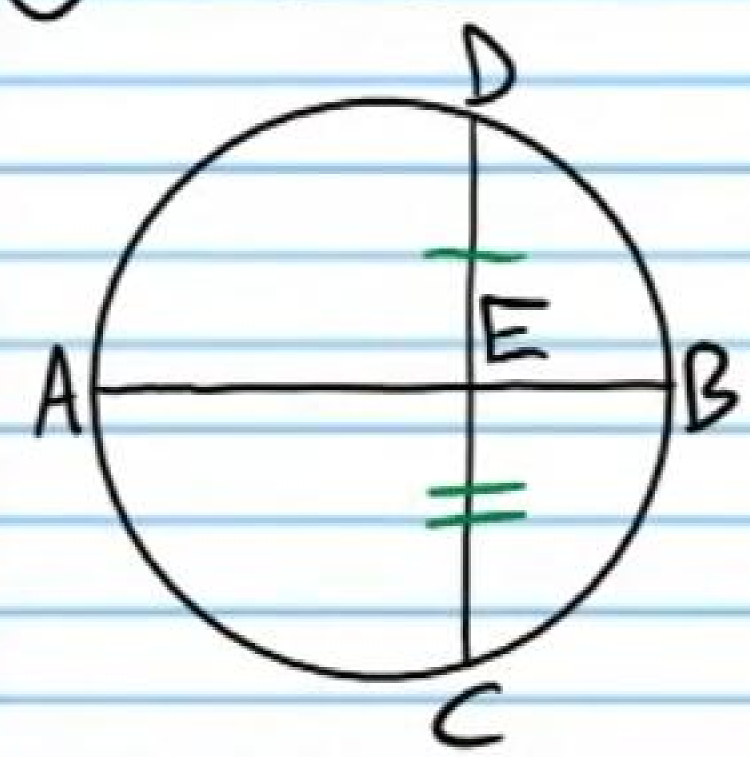
Yes, $CE = ED$ & $\overline{AB} \perp \overline{CD}$, so $\overline{AB}$ is a diameter.

11.



$DH = HE$, but we do not know that $\overline{AB} \perp \overline{DE}$, so we cannot conclude that $\overline{AB}$ is a diameter.

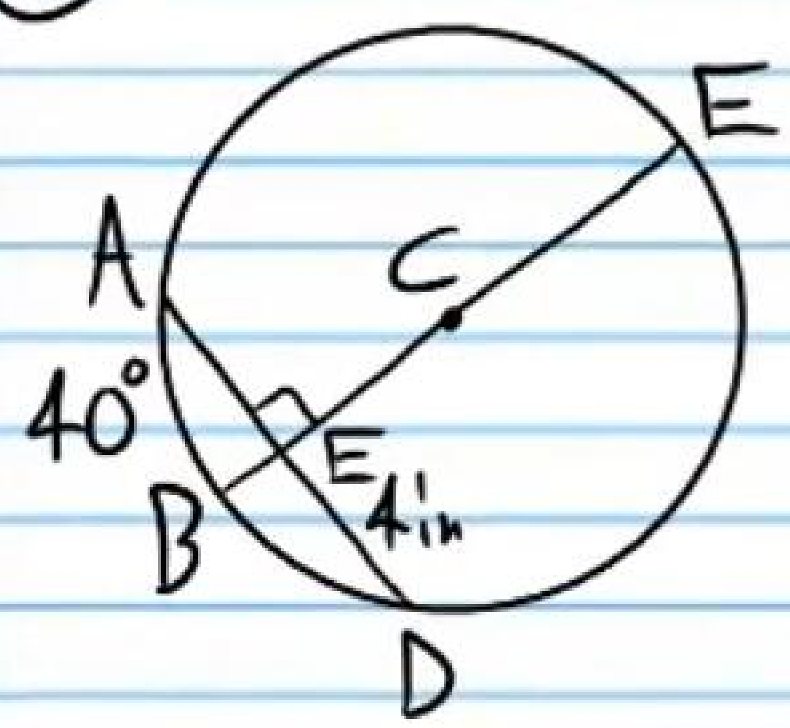
12. $m\angle DEB = 90^\circ$



$\overline{AB} \perp \overline{DC}$, but $DE \neq EC$, so we cannot conclude that $\overline{AB}$ is a diameter.

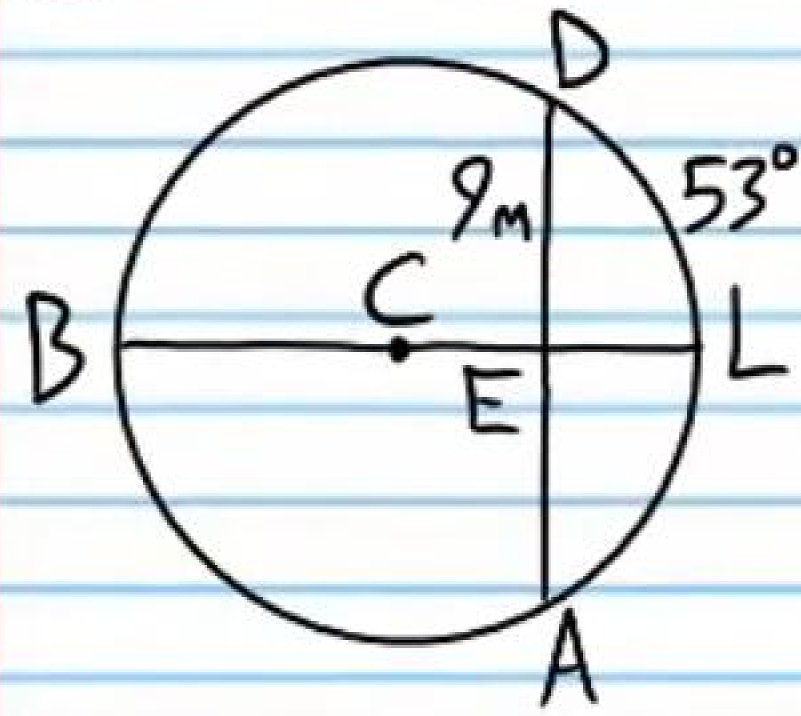
If C is the center of each circle below, find the following values.

(13) AE & $m\widehat{BD}$



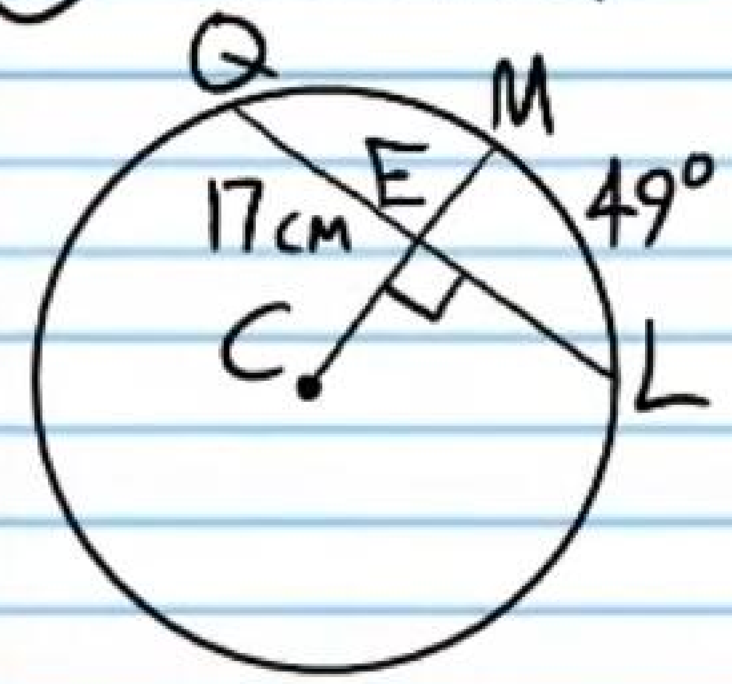
$AE = 4$ in
$m\widehat{BD} = 40°$

(14) $m\widehat{AD}$ & AD



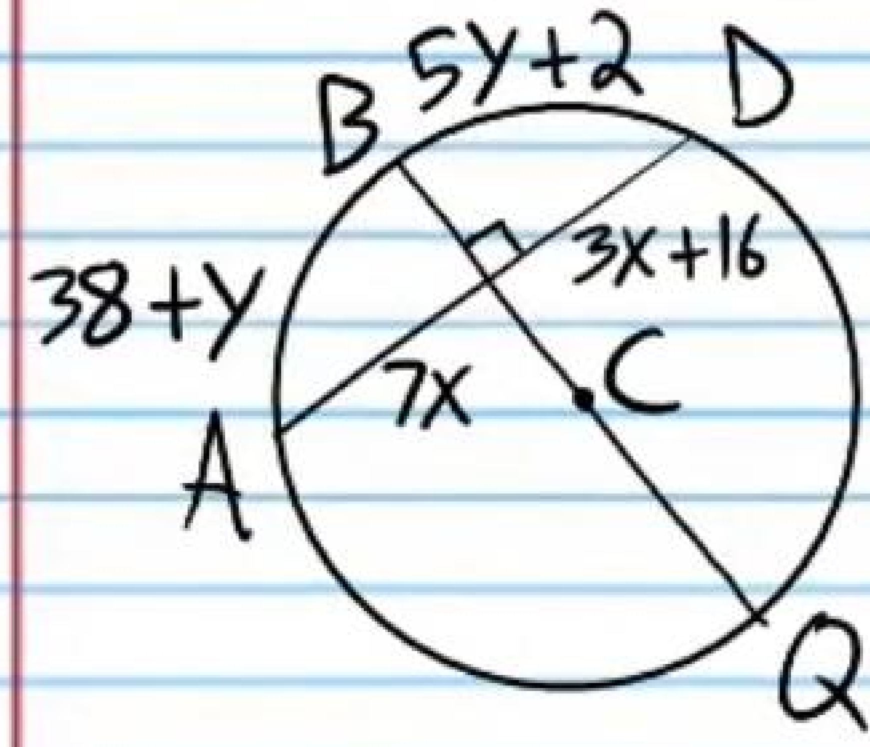
We do not know that $\overline{BL} \perp \overline{AD}$, so we cannot use the $\perp$ Diameter Theorem.

(15) QL & $m\widehat{QL}$



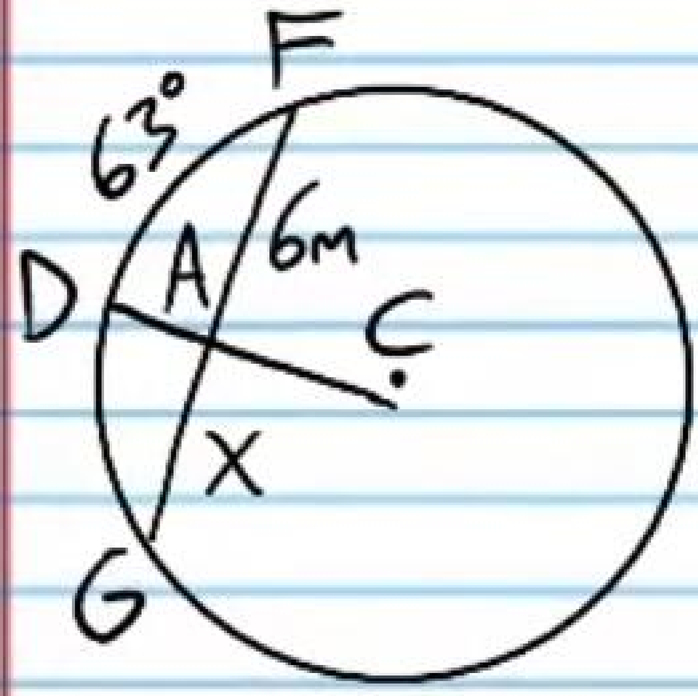
$QL = 34$ cm
$m\widehat{QL} = 98°$

(16) AD & $m\widehat{AB}$



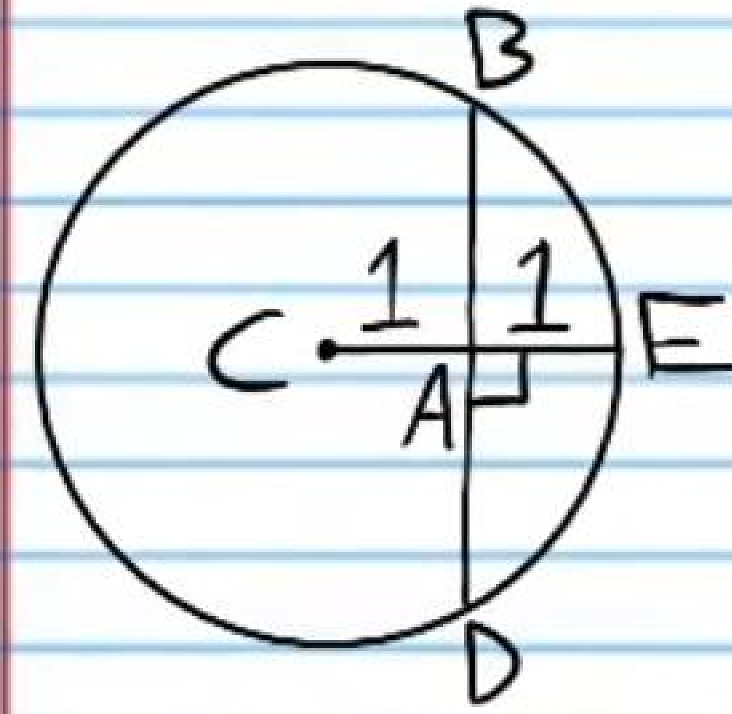
$m\widehat{AB} = 47°$
$AD = 56$

(17) $m\widehat{DG}$ & FG



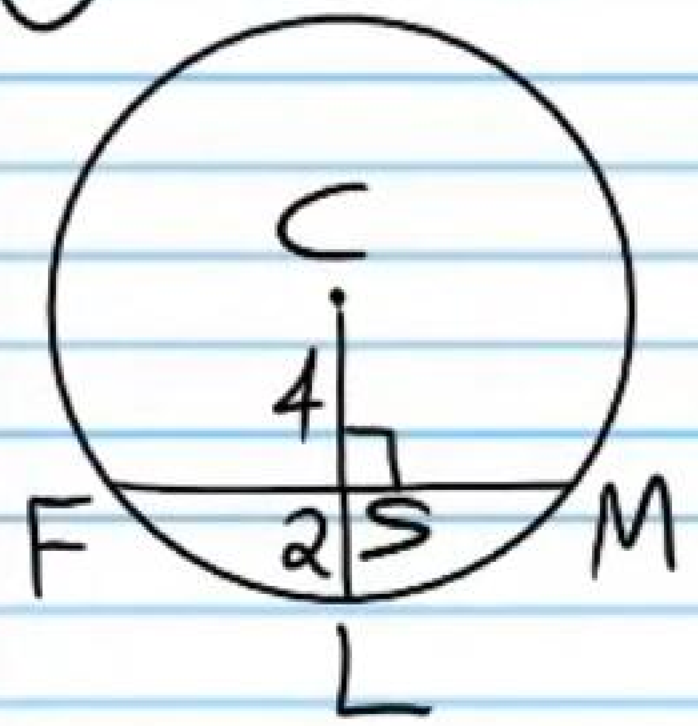
$\overline{DA}$ is not a radius, so we cannot use the $\perp$ Diameter Theorem.

(18) BD



$\boxed{2\sqrt{3}}$

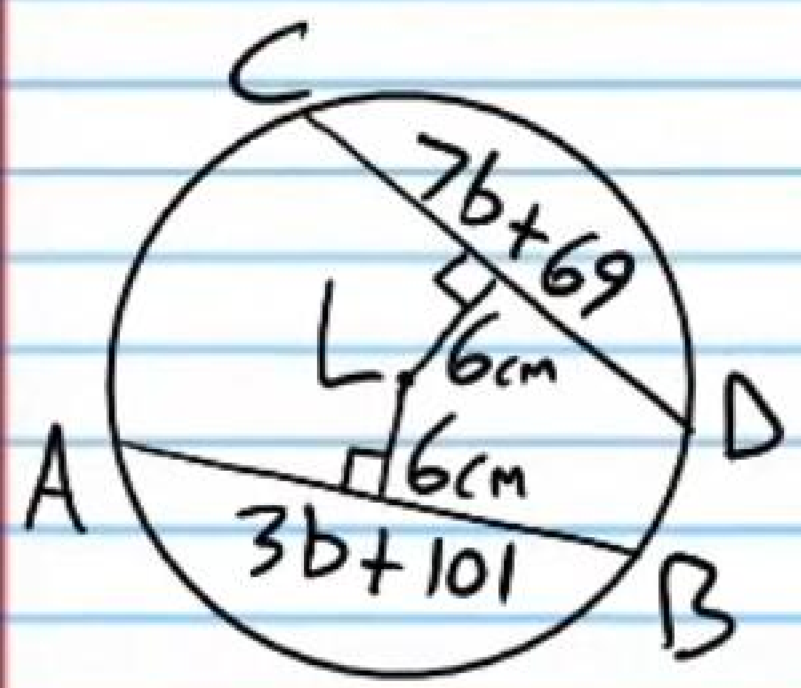
(19) FS



$\sqrt{20} = \boxed{2\sqrt{5}}$

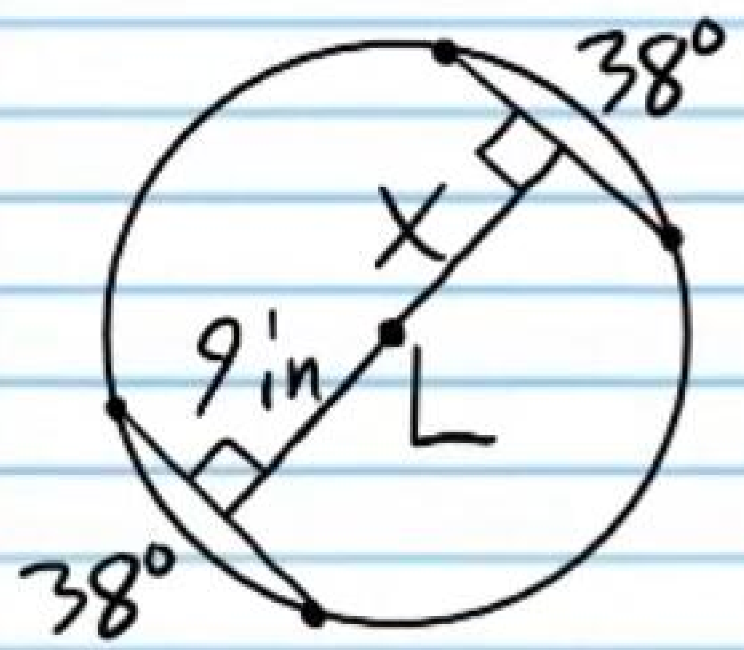
If L is the center of each circle below, find the following values.

(20) AB



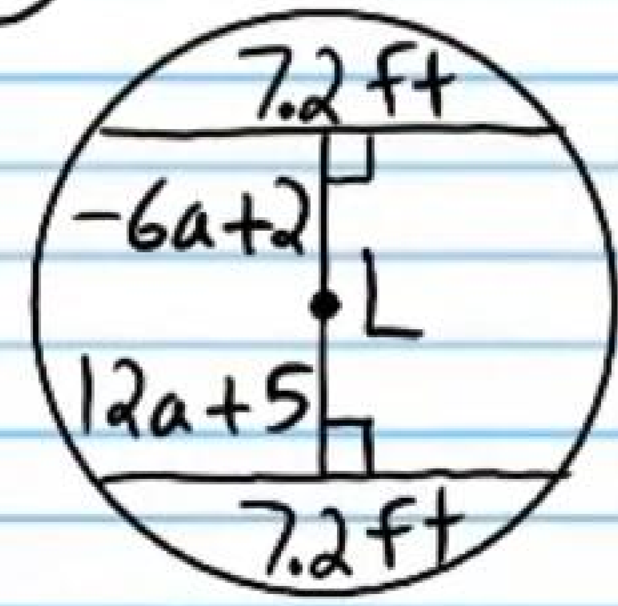
$\boxed{125}$

(21) X



$\boxed{9\text{ in}}$

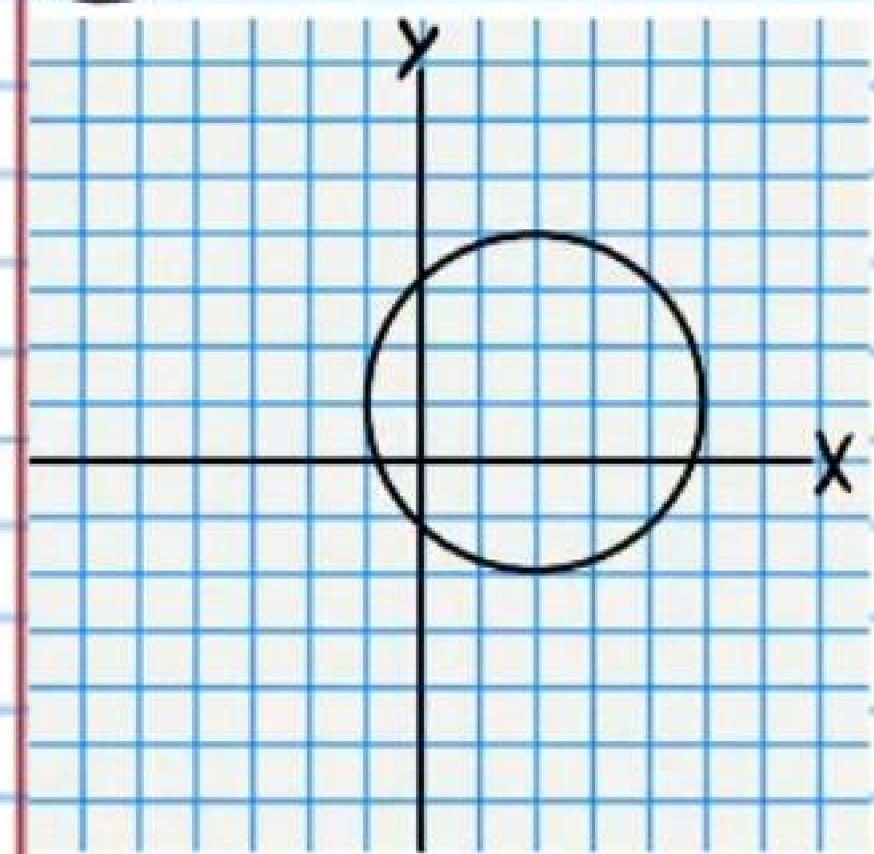
(22) a



$\boxed{-\frac{1}{6}}$

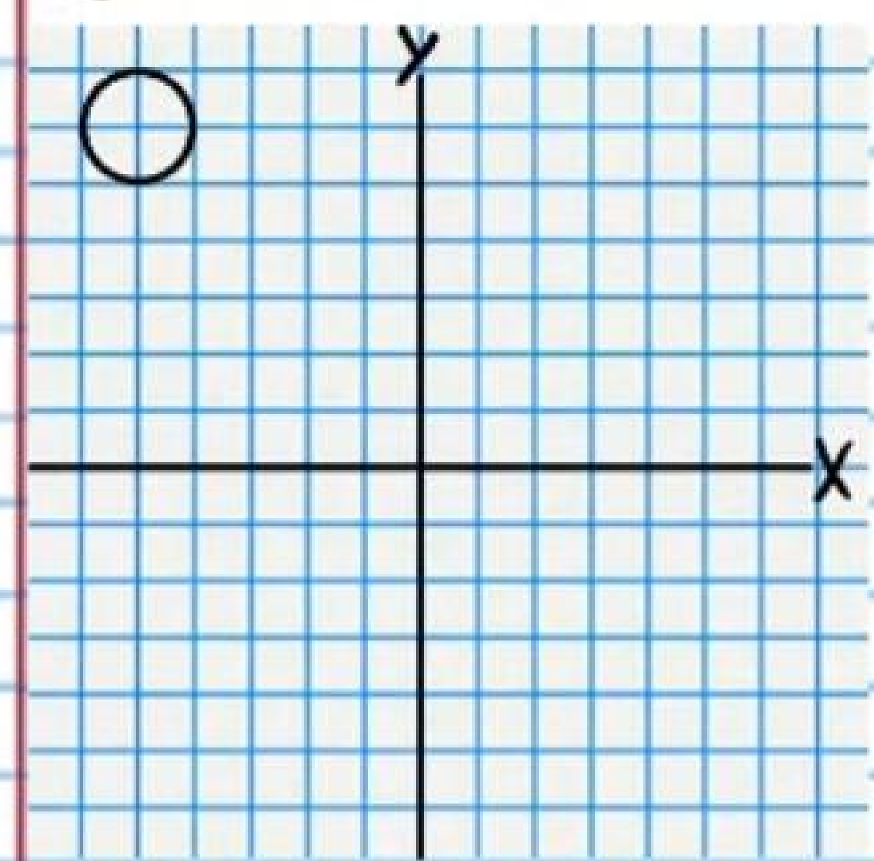
Write the equations of the following circles.

(23)



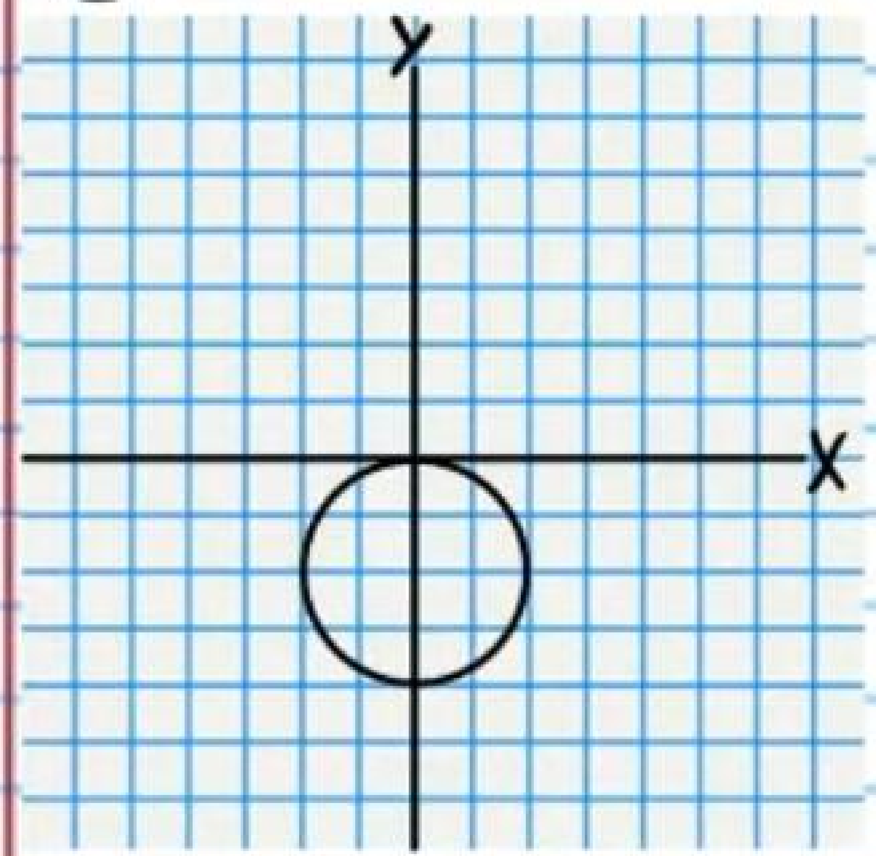
$\boxed{(x-2)^2+(y-1)^2=9}$

(24)



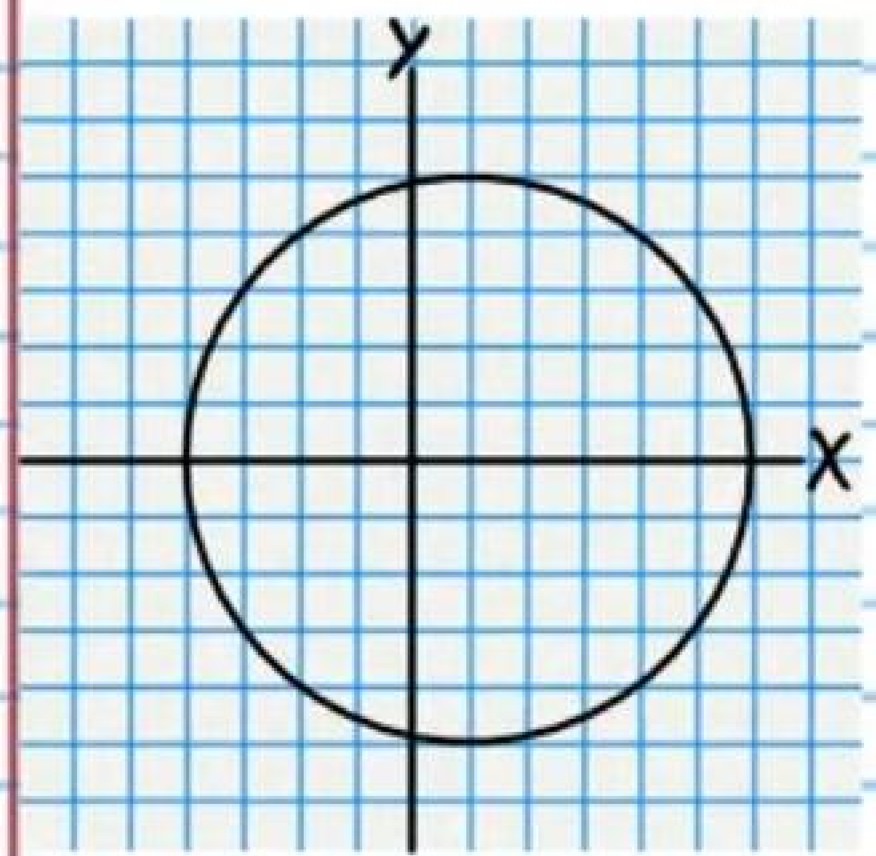
$\boxed{(x+5)^2+(y-6)^2=1}$

(25)



$\boxed{x^2+(y+2)^2=4}$

(26)



$\boxed{(x-1)^2+y^2=25}$

Graph the equations below. * Use graph paper.
Label axes.

(27) $(x+3)^2+(y+3)^2=9$

(28) $(x+2)^2+(y-1)^2=1$

(29) $x^2+y^2=4$

(30) $(x-3)^2+(y+2)^2=16$

Write the equations of the circles with the given characteristics.

(31) Center: $(-2,1)$
Point on the circle: $(3,3)$ $\boxed{(x+2)^2+(y-1)^2=29}$

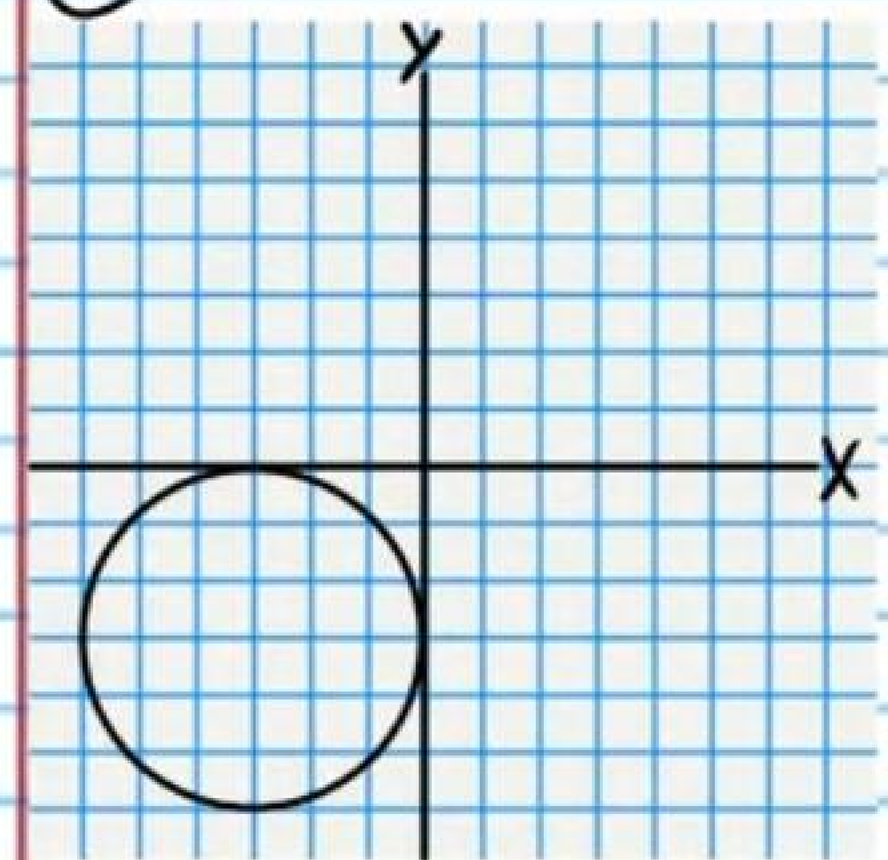
(32) Center: $(-3,-4)$
Point on the circle: $(-1,-5)$ $\boxed{(x+3)^2+(y+4)^2=5}$

(33) Diameter with endpoints $(0,4)$ & $(6,6)$. $\boxed{(x-3)^2+(y-5)^2=10}$

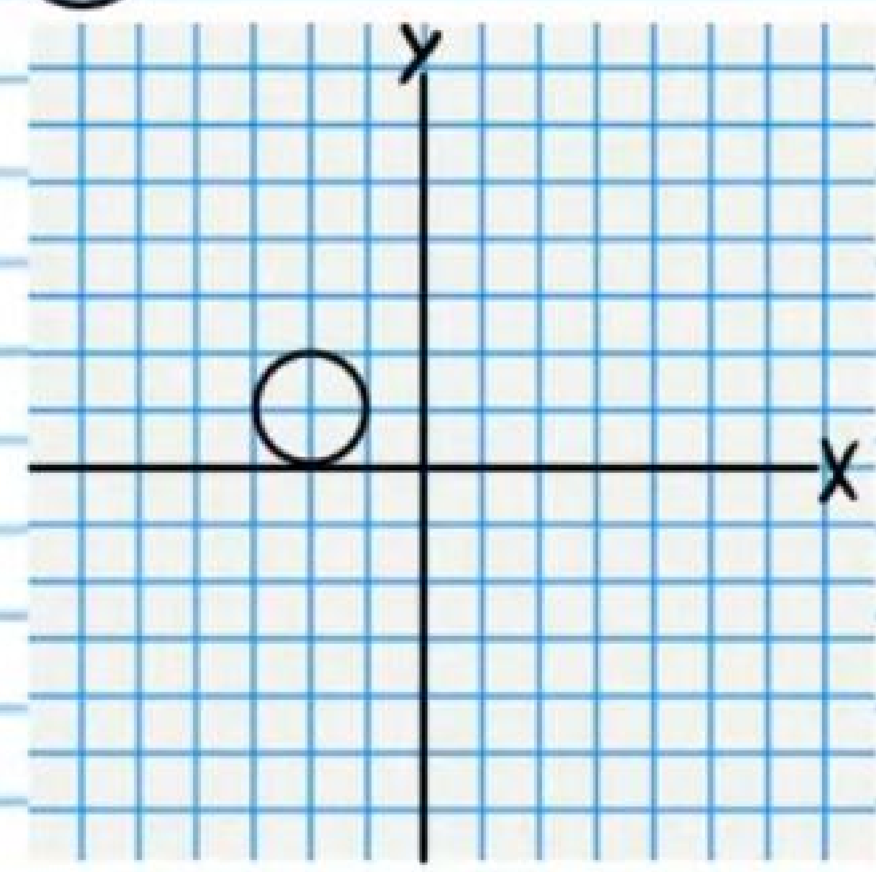
(34) Diameter with endpoints $(-2,6)$ & $(1,5)$. $\boxed{\left(x+\frac{1}{2}\right)^2+\left(y-\frac{11}{2}\right)^2=2.5}$

Answers for #'s 27-30.

(27)



(28)



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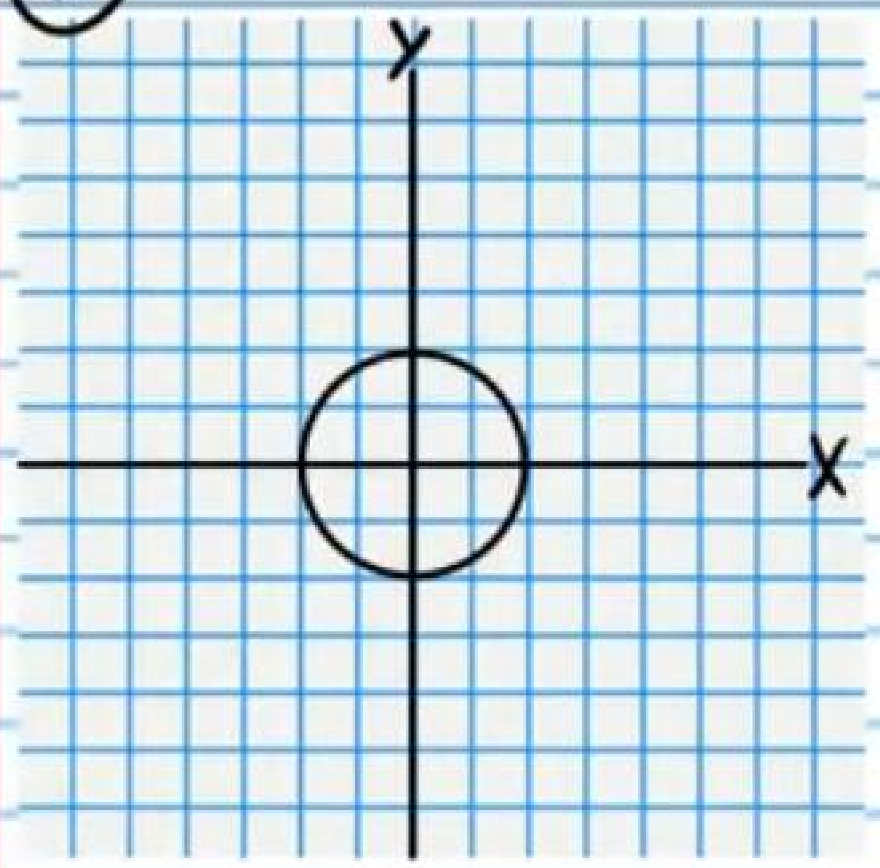

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